AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 9 · STUDENT LESSON

Prompt Evaluation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 28: compare prompts on the same inputs and criteria.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of controlled prompt experiments. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of controlled prompt experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

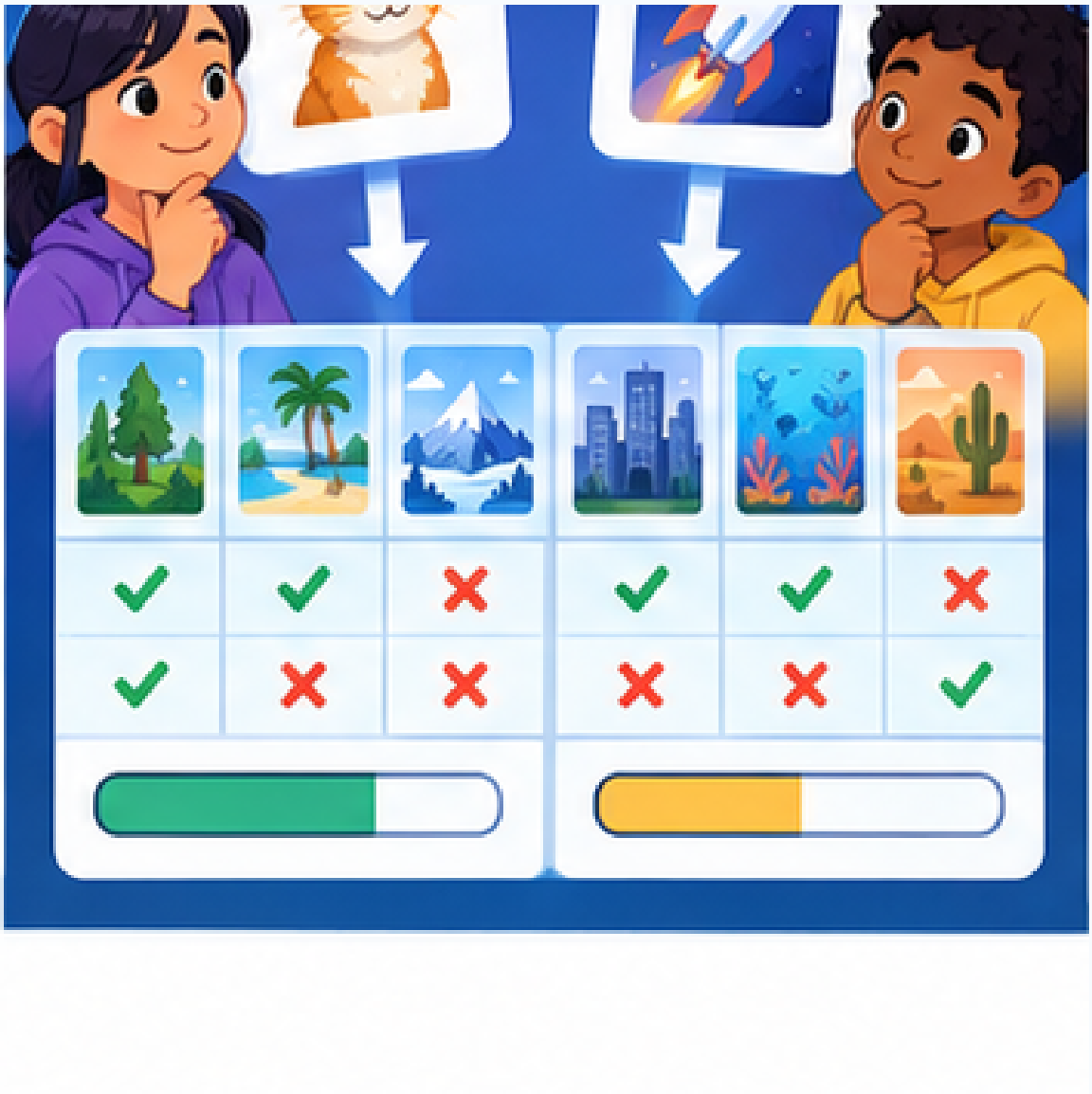
Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 9: Prompt Evaluation

Prompt claims require a fixed **test set** and repeatable scoring.



CREATOR MISSION

A/B test two prompts across six cases.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Structured **output**s and validation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can evaluate prompts with **paired case**s and a predeclared mandatory criterion.

STUDENT LESSON · 2 OF 18

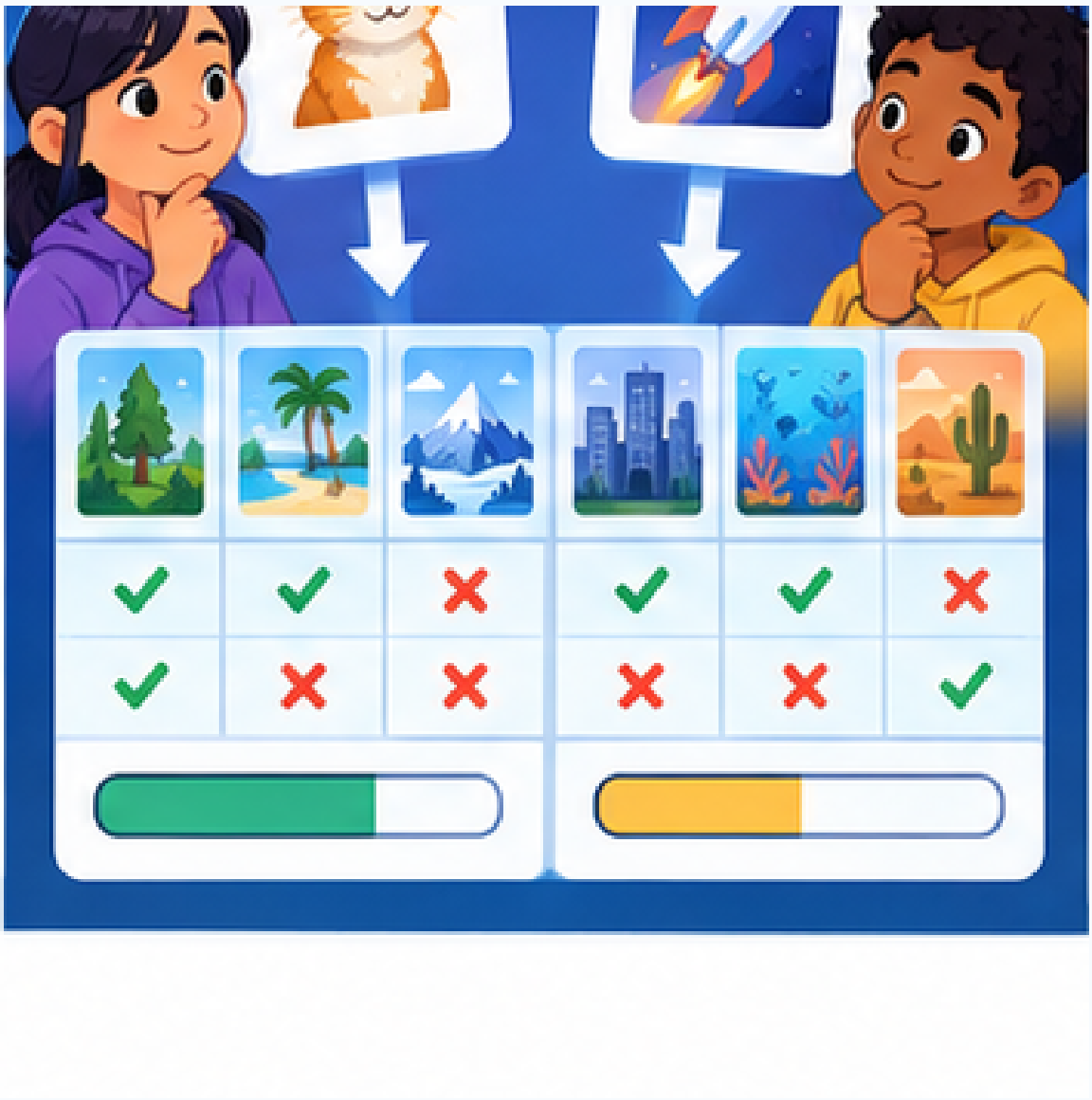
Notice the evidence

> Investigate the supplied Prompt Evaluation case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use prompt evaluation to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

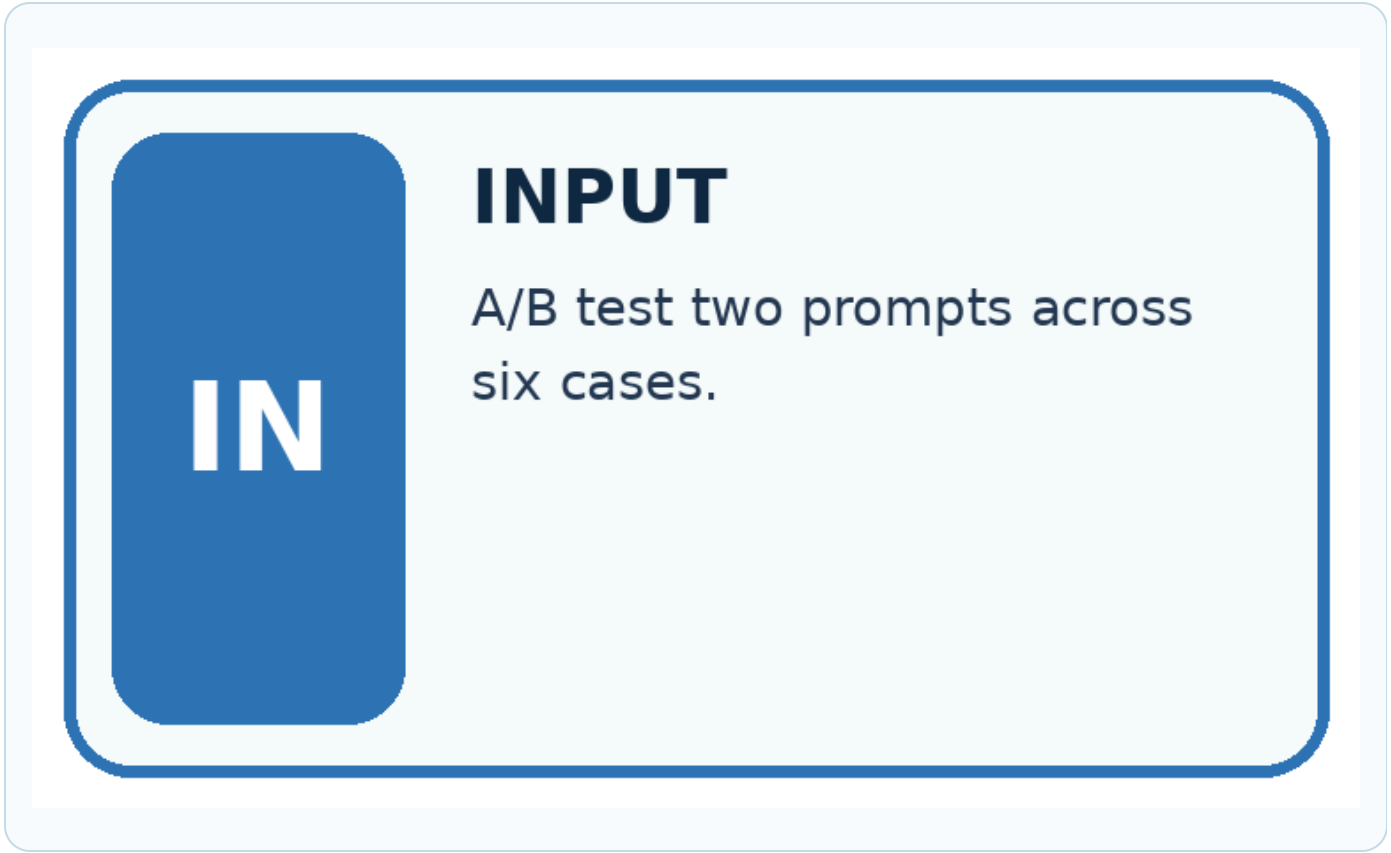
See how it works

> Follow the mechanism for Prompt Evaluation. Connect each labelled input, deciding step and result.

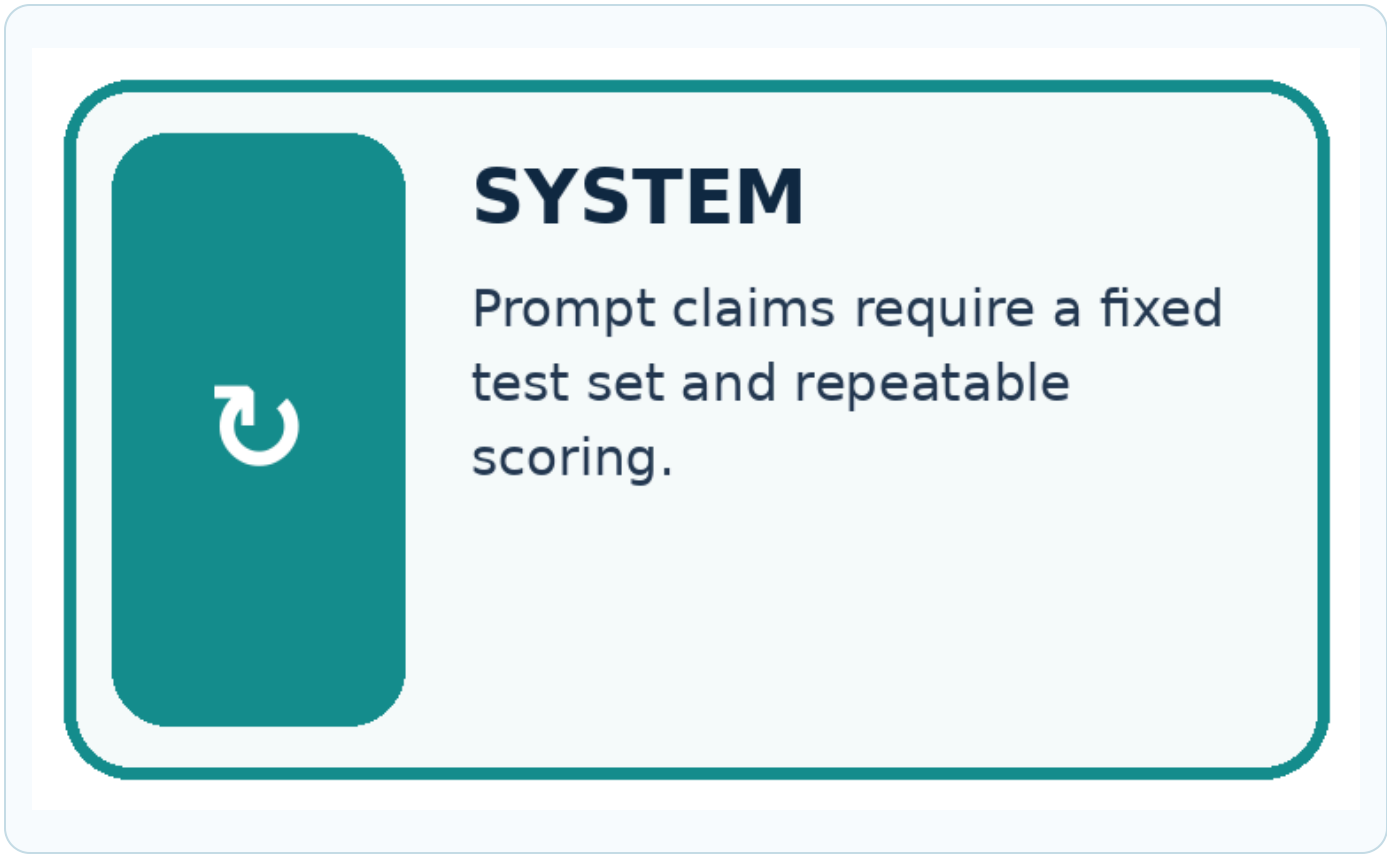
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Prompt claims require a fixed test set and repeatable scoring.
OUT
EVIDENCE
A/B test two prompts across six cases.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Evaluate both prompts on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

Controlled prompt experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.	Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.	P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

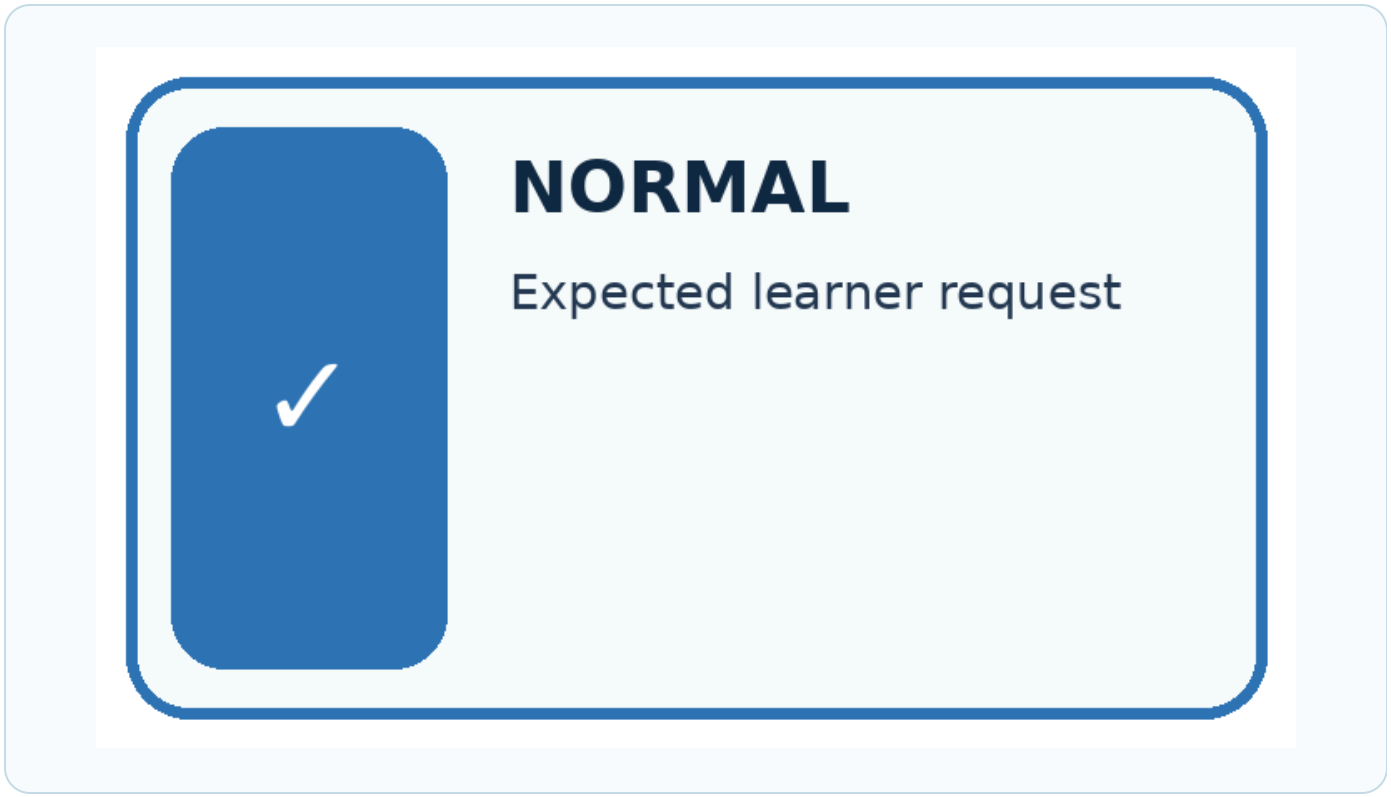
> Compare the supplied cases for Prompt Evaluation. Identify the detail that changes the conclusion.

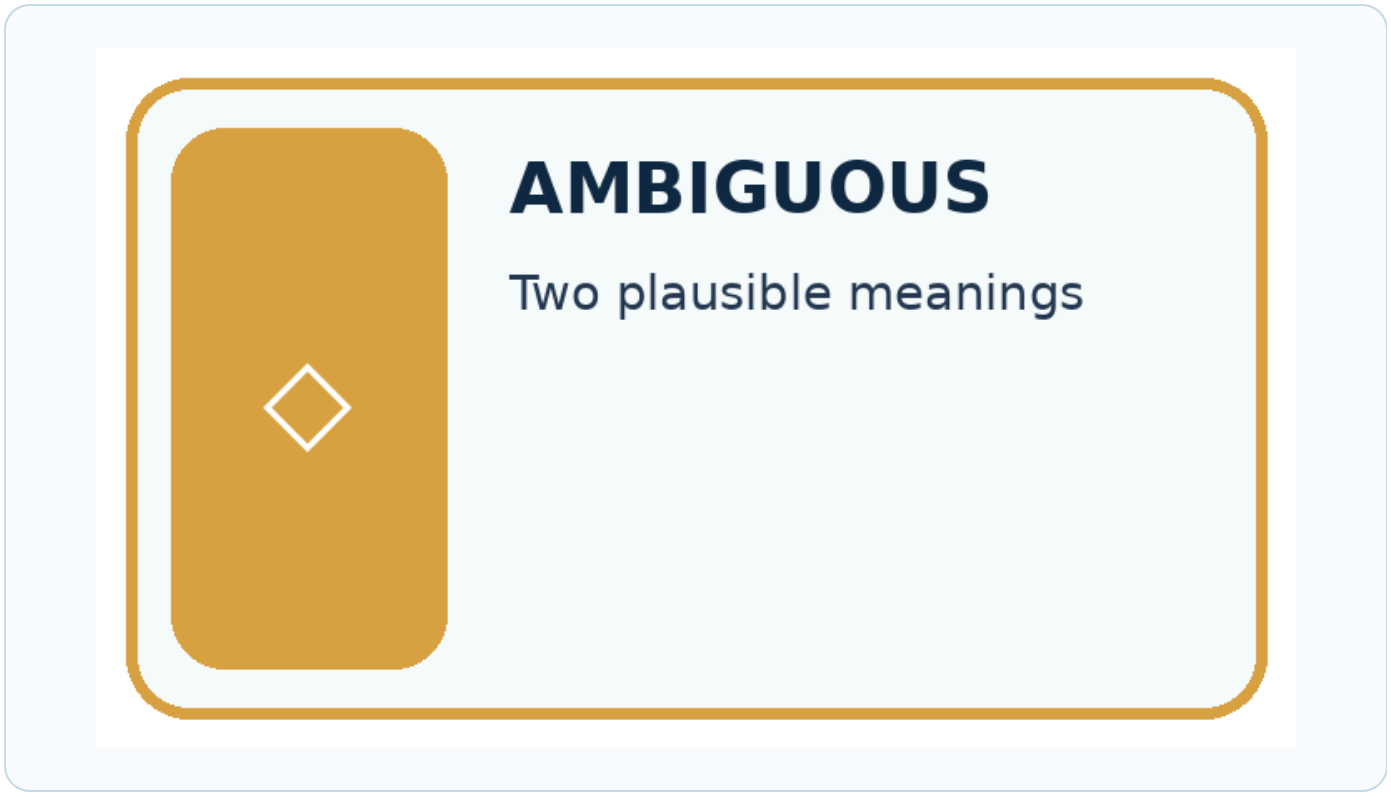
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Remove the gate and retain total passes as the only criterion.

Predicted result

Prediction: The record is a tie; another predeclared tie policy or further development **evidence** is needed.

Why the result follows

Reason: Both prompts have four passes on six questions.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Prompt Evaluation to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

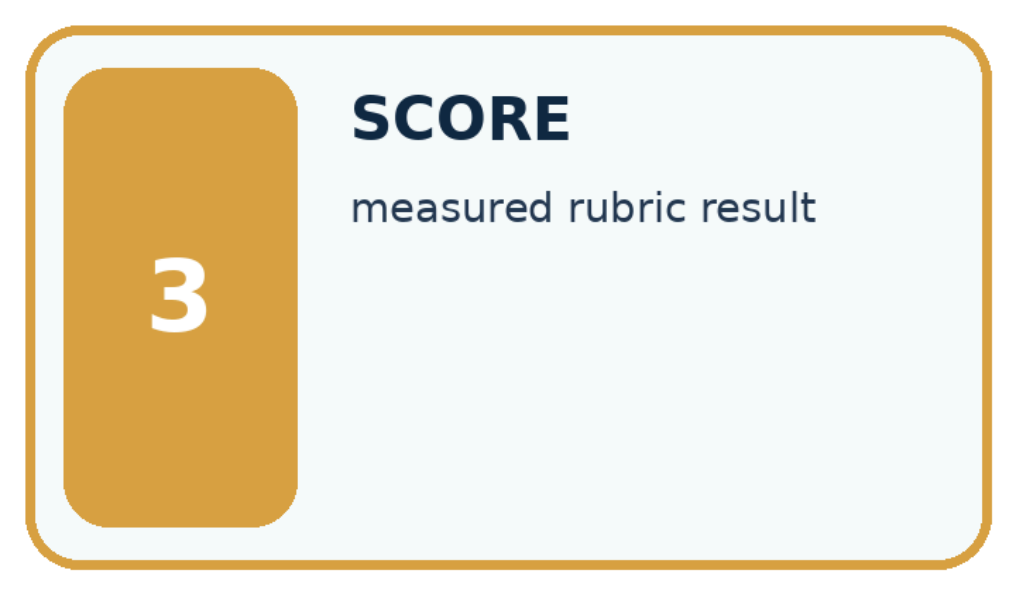
A/B TEST

compare two versions under the same cases

2

TEST SET

fixed examples used for evaluation





Word	My own example	Not this / boundary
A/B test	A comparison of two versions using a planned assignment of comparable cases and the same outcome measure.	
test set	Data reserved for evaluating a selected system; final test data must not influence its development.	

Word	My own example	Not this / boundary
score	A numerical value assigned using a specified rule; it is not automatically a probability.	
variance	A measure of spread around a mean; informally, variation across repeated results. Specify which meaning is intended.	

Words used in the closer explanation

Word	Meaning
Paired case	The same case evaluated under each compared procedure
Rubric	Prespecified criteria for judging an output
Regression	A previously passing condition that fails after a change

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STUDENT LESSON · 6 OF 18

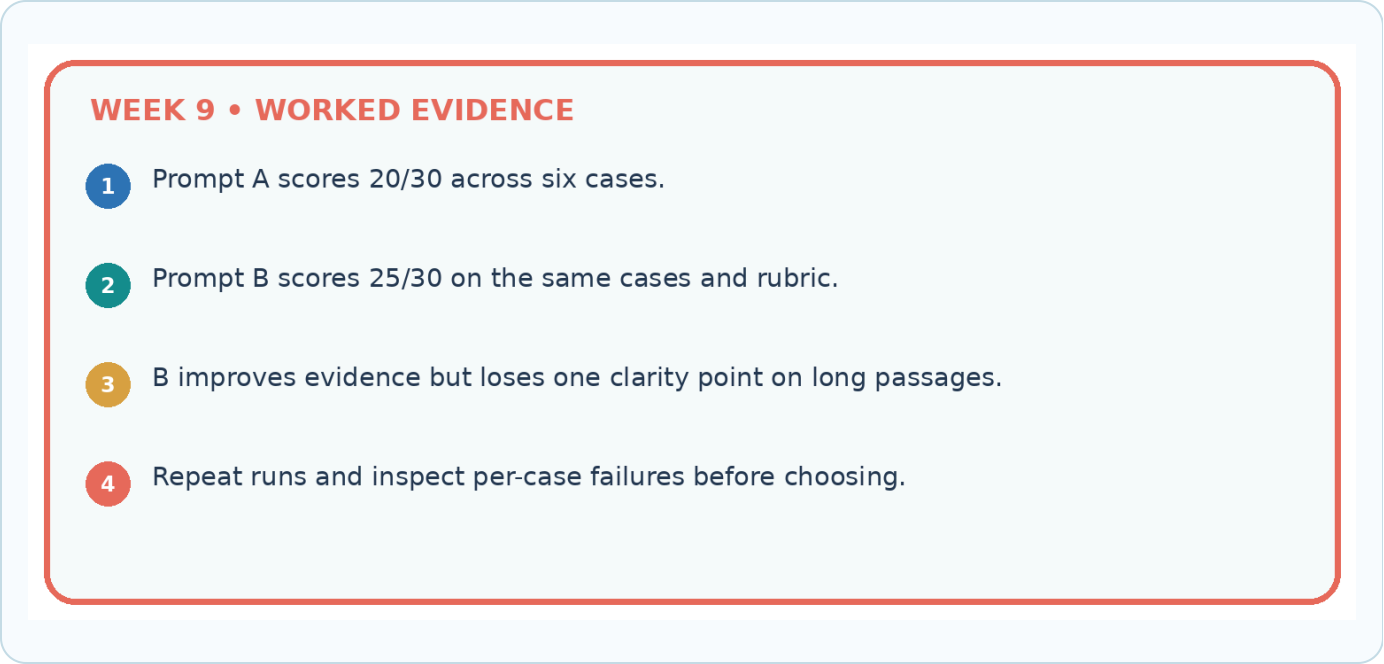
Follow a worked example

> Follow the worked trace for Prompt Evaluation, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Prompt A **score**s 20/30 across six cases.

2. Prompt B **score**s 25/30 on the same cases and **rubric**.

3. B improves **evidence** but loses one clarity point on long passages.

4. Repeat runs and inspect per-case failures before choosing.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Worked explanation · Prompt Evaluation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompt Evaluation

A fictional Study Buddy compares two prompts across six development questions.

The facts we will use

Supplied facts

A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Supplied facts

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

AI ACADEMY / AI-3/9

Same total, different mandatory-case results

	Total passes /6	Refusal passes /2
Prompt A	4	0
Prompt B	4	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/9

Prompt Evaluation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired total

A and B both pass 4/6.

↓



2. Different failures

A alone passes 3,4; B alone passes 5,6.

↓



3. Mandatory slice

A passes 0/2 refusal cases; B passes 2/2.

↓



4. Selection

B clears the declared gate, but its failures 3,4 remain visible.

CHANGE ONE THING

Remove the gate and retain total passes as the only criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Paired total	A and B both pass 4/6.
2. Different failures	A alone passes 3,4; B alone passes 5,6.
3. Mandatory slice	A passes 0/2 refusal cases; B passes 2/2.
4. Selection	B clears the declared gate, but its failures 3,4 remain visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

What this example does not establish

Keep the stated limit

Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

Keep the stated limit

Read the labelled picture: Why is B's selection defensible here without calling it universally superior?

Keep the stated limit

Change one thing: Remove the gate and retain total passes as the only criterion. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Prompt Evaluation claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A single impressive output proves one prompt is best. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

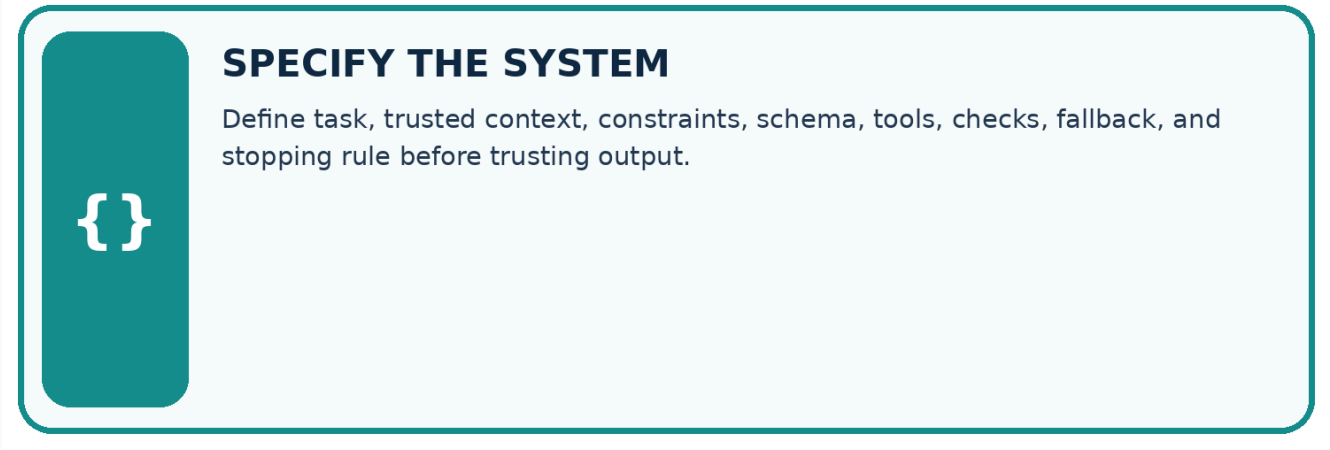
Practise together

Practise Prompt Evaluation with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Prompt selection on development cases cannot be reported as independent final performance.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled prompt experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is B’s selection defensible here without calling it universally superior?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Prompt Evaluation. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Freeze six cases, run A and B with one changed variable, **score** blind with the same **rubric**, and explain wins and **regression**s.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

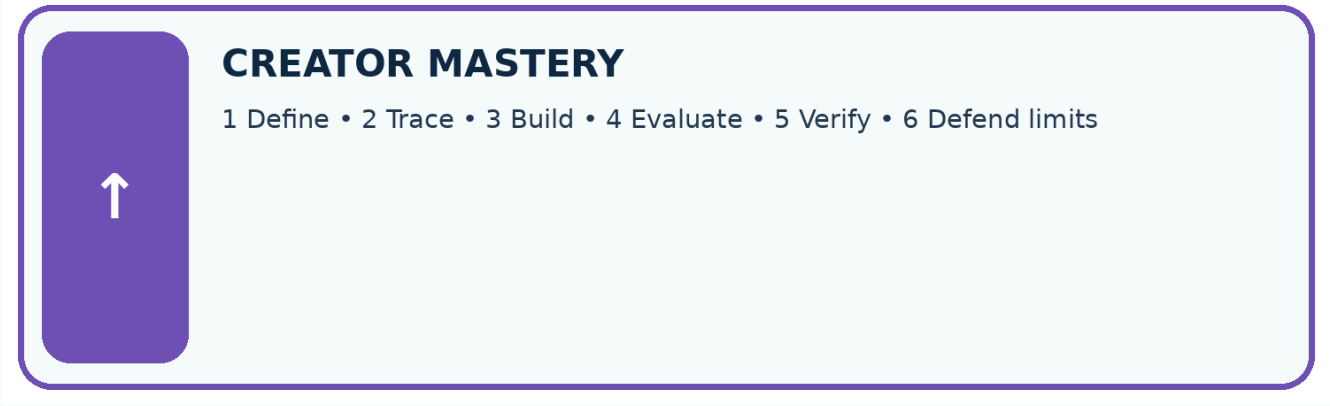
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Prompt Evaluation. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

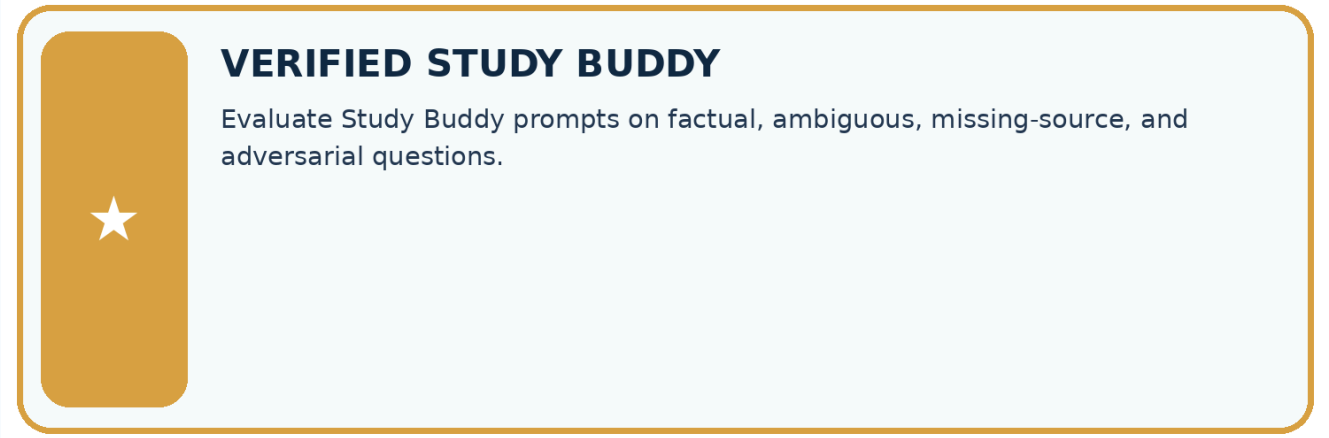
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

STUDENT LESSON · 12 OF 18

Show what you know

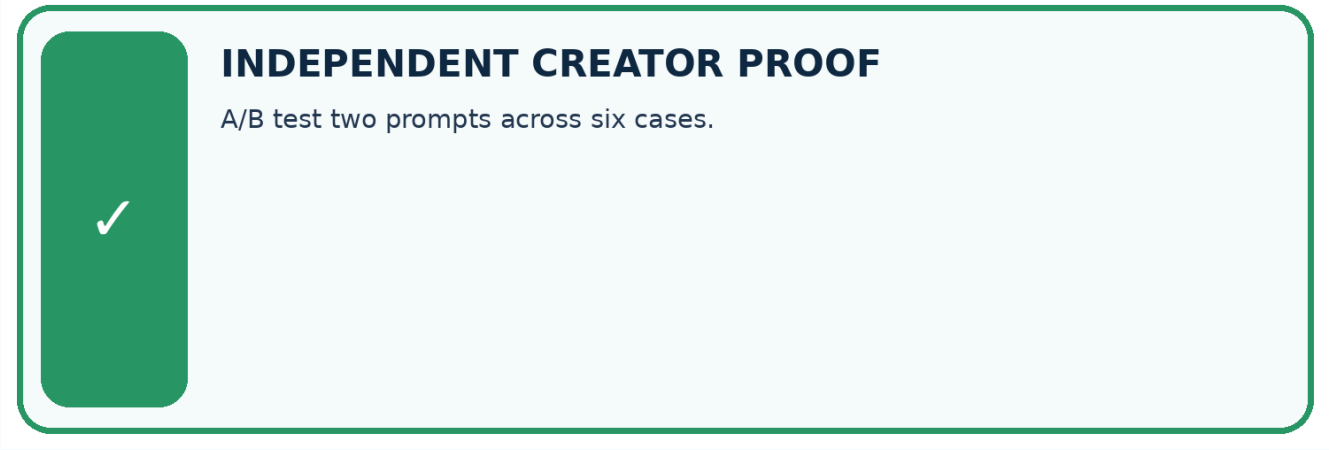
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE **A/B test** two prompts across six cases.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Prompt Evaluation case for supported practice. Repeated examples do not become fresh assessments.

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

On five fixed questions, A passes four and B passes three; B looks more attractive.

The question

> What conclusion is supported?

Worked explanation

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Evaluate prompts on the same questions

Evidence or stage	Value or result
Prompt A	4/5
Prompt B	3/5

On five fixed questions, A passes four and B passes three; B looks more attractive.

What conclusion is supported?

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Five validation tasks have pass results A:1,1,1,0,0 and B:0,1,0,1,1. Compare performance and explain what is required before choosing a prompt.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A passes 8 of 10 tasks; B passes 9 of 10 but fails the only source-checking task. A

source check is mandatory. Which can be accepted as-is?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

On four shared tasks, A **score**s 1,1,0,1 and B **score**s 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Prompt Evaluation, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional Study Buddy compares two prompts across six development questions.

By the end, I can	How I will show it
Explain the mechanism and terminology of controlled prompt experiments	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Prompt Evaluation without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of controlled prompt experiments

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy compares two prompts across six development questions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B clears the declared gate, but its failures 3,4 remain visible.

1. Different failures

2. Paired total

3. Mandatory slice

4. Selection

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Selection B clears the declared gate, but its failures 3,4 remain visible. Equal totals do not imply equal behavior.

The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result. Selecting B is justified under that rule, not by claiming it is best on every question.

After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove the gate and retain total passes as the only criterion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The record is a tie; another predeclared tie policy or further development **evidence** is

needed.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Both prompts have four passes on six questions.

Repair a missing building block

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Examples and **Rubric**s (ai-3/6). Goal: Write a prompt specification with observable response criteria. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card..

Return to this week and explain what you changed.

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

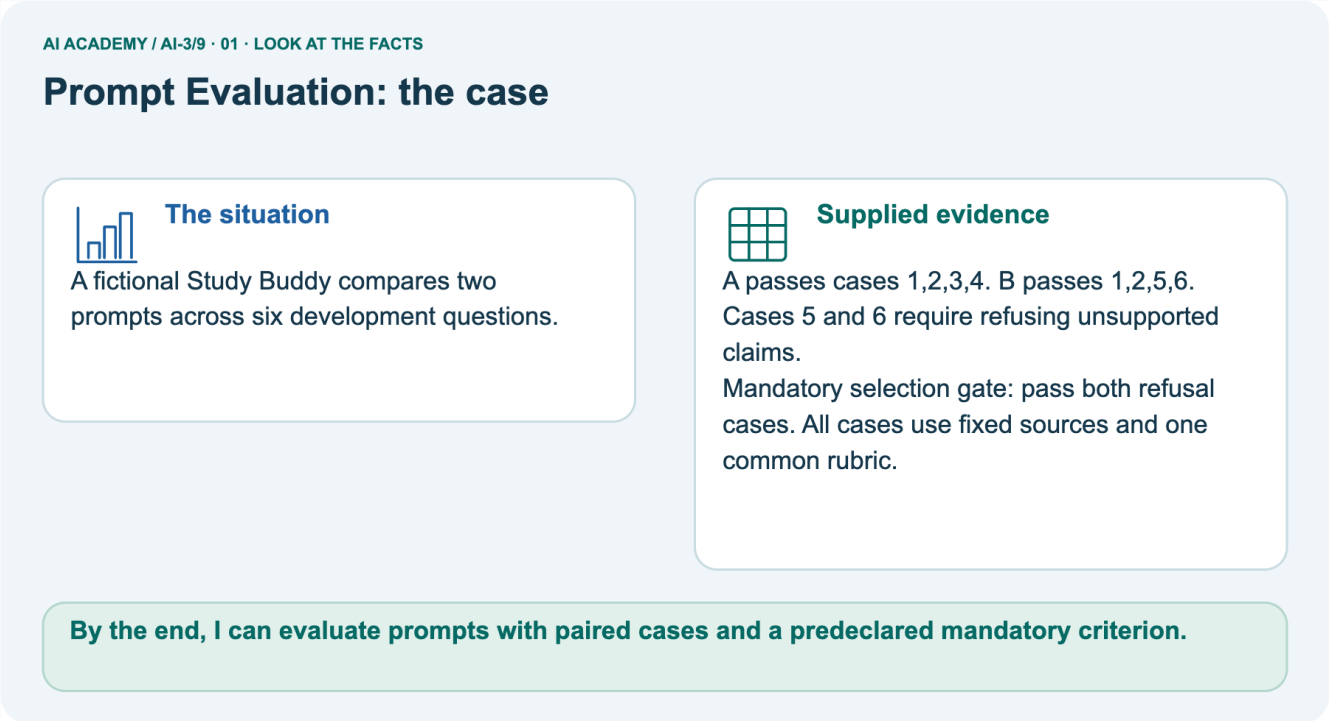
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Prompt Evaluation graphic using its labels, deciding evidence and relevant limitation.

01. Prompt Evaluation: the case



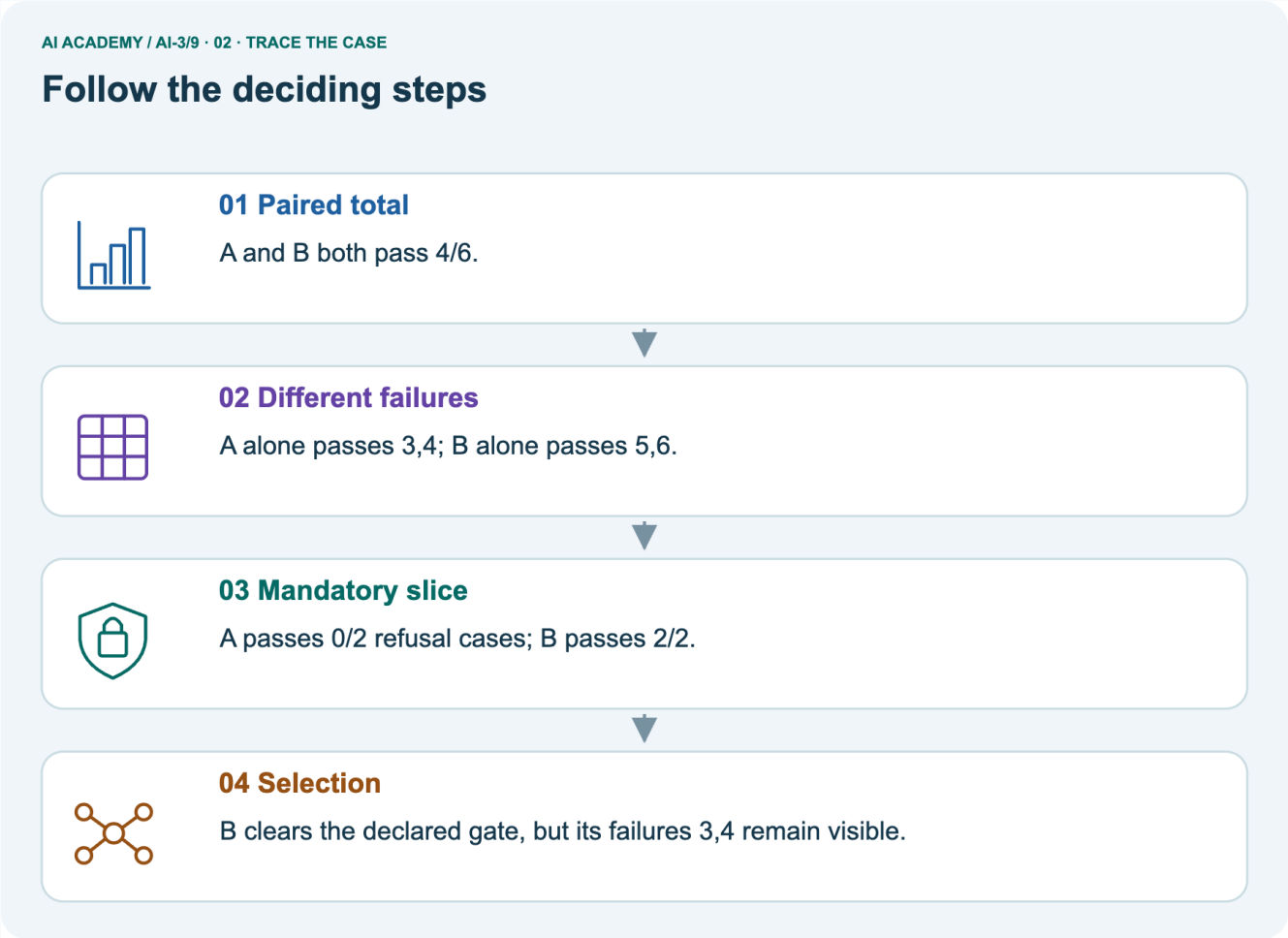
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy compares two prompts across six development questions. Supplied facts: A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

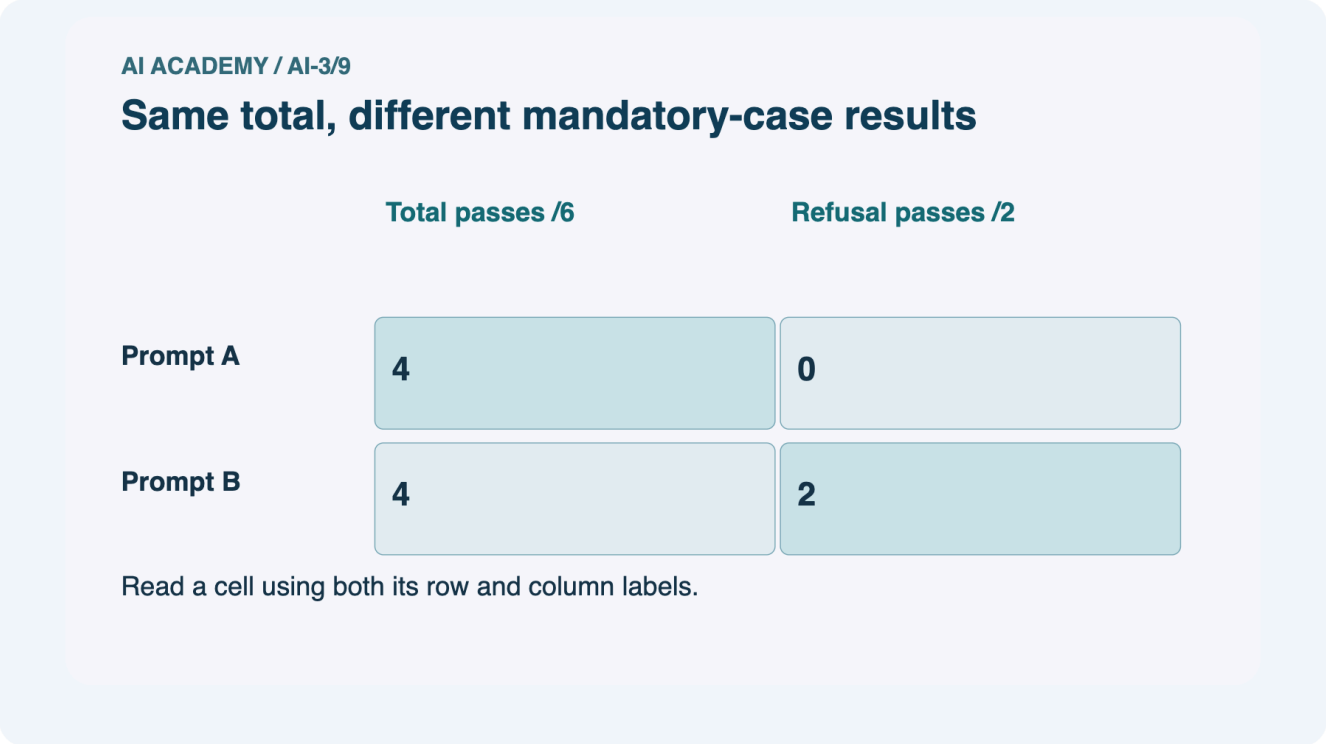


Follow the labelled picture with the matching explanation. Open the vector diagram

Paired total: A and B both pass 4/6. → Different failures: A alone passes 3,4; B alone passes 5,6. → Mandatory slice: A passes 0/2 refusal cases; B passes 2/2. → Selection: B clears the declared gate, but its failures 3,4 remain visible.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Same total, different mandatory-case results

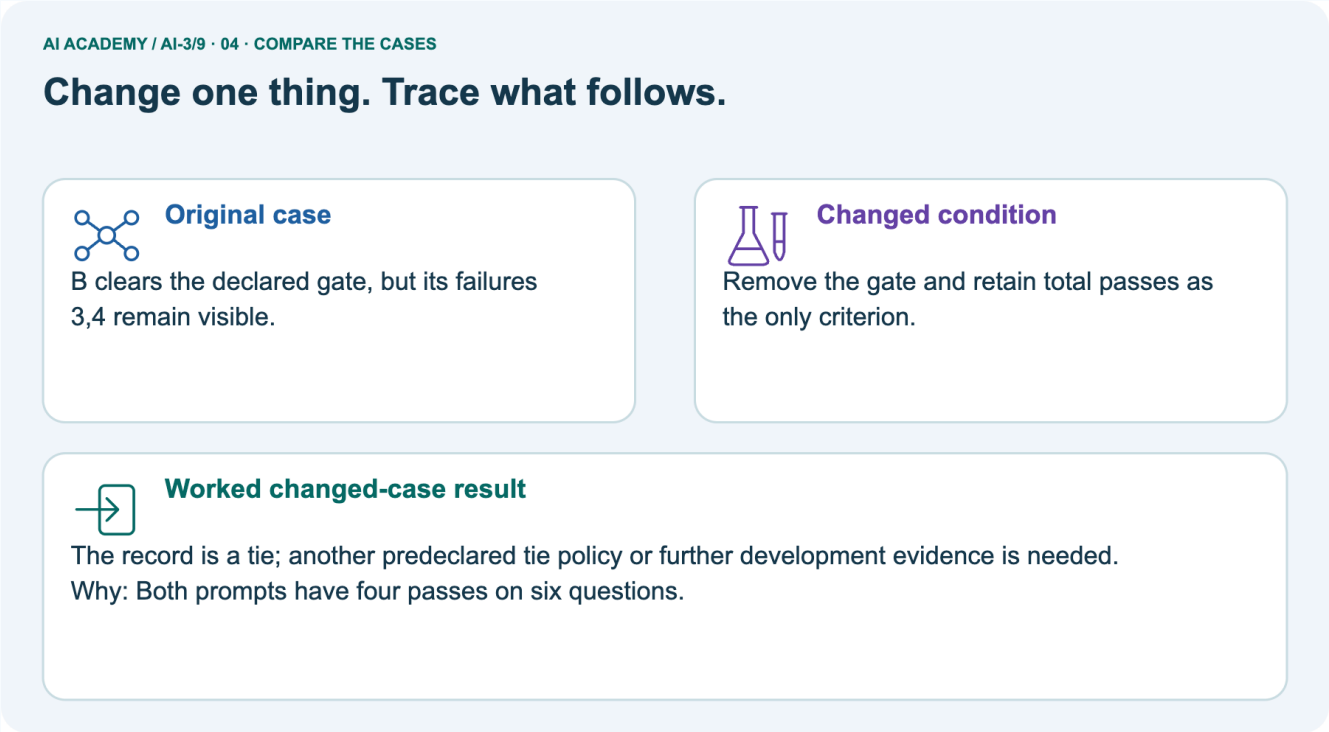


Follow the labelled picture with the matching explanation. Open the vector diagram

Same total, different mandatory-case results. Prompt A: Total passes /6 = 4, Refusal passes /2 = 0. Prompt B: Total passes /6 = 4, Refusal passes /2 = 2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



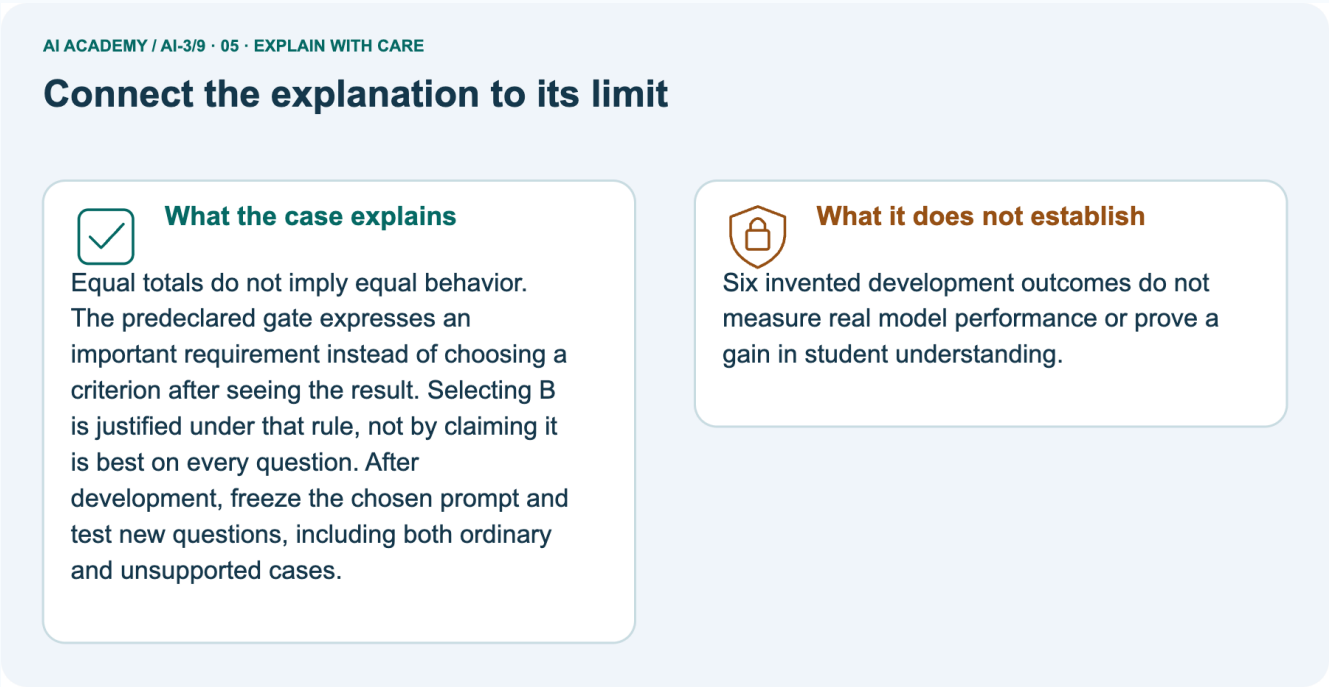
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B clears the declared gate, but its failures 3,4 remain visible. Changed condition: Remove the gate and retain total passes as the only criterion.

Expected result: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Reason: Both prompts have four passes on six questions.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

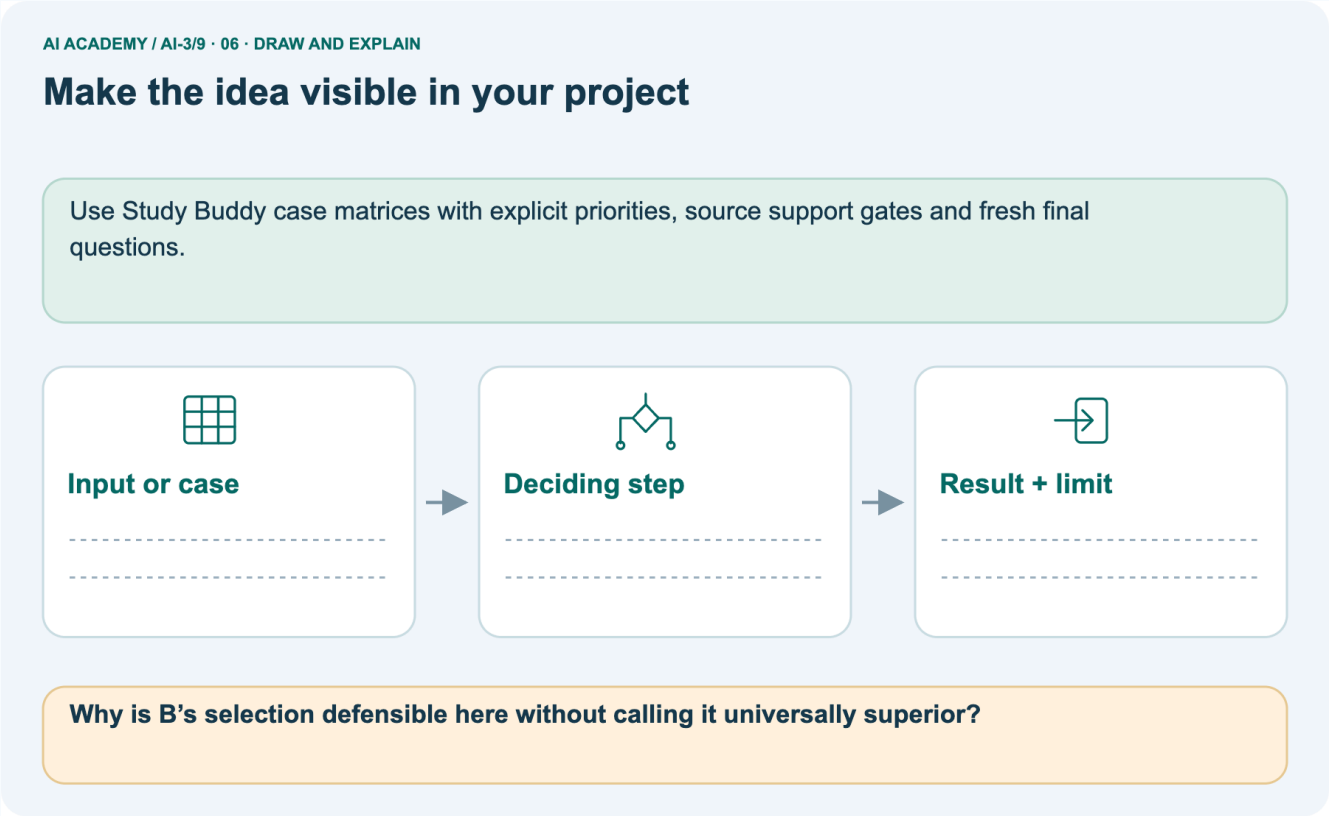
Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Limit: Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions. Draw your **input**, deciding step, result and limit.

Supported question: Why is B's selection defensible here without calling it universally superior?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.